

Predictors of outcome in pediatric epilepsy surgery

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Article abstract—*Objective:* To determine the correlation between pre- and perioperative variables on the outcome of children undergoing focal resections for medically intractable partial epilepsy. *Methods:* Retrospective analysis of pre- and perioperative variables in a cohort of 75 patients younger than 12 years of age who underwent excisional surgery and had at least 1 year of follow-up. Outcome, measured by postoperative seizure frequency, was analyzed as a function of age at seizure onset, duration of epilepsy, presence of cognitive impairment, lobe of seizure origin, presence of a lesion, histopathology, and completeness of resection. Completeness of resection was defined on the basis of excising both the entire structural lesion if present and the region revealing prominent interictal and ictal abnormalities on intracranial EEG. *Results:* Seventy-seven percent of patients had good outcomes (class 1 or 2), and 59% were seizure-free. Lesional status, site of resection, and pathologic diagnoses were not significant predictors of outcome except for in multilobar resection, for which overall outcome was relatively poor (44% class 3 or 4; 22% seizure-free). Completeness of resection was the only significant predictor of good outcome ($p < 0.001$), with 92% of patients who underwent complete resection of the epileptogenic zone achieving good outcome compared with 50% of patients who had incomplete resections. *Conclusion:* In this series of pediatric patients, complete resection of the lesion and the electrographically abnormal region was the main determinant of outcome after focal resections. Except for multilobar resections, other factors examined in this study did not significantly influence postoperative seizure prognosis and should not influence candidate selection for the surgical process. **Key words:** Epileptogenic zone—Focal epilepsy—Nonlesional temporal lobe epilepsy—Pediatric epilepsy—Surgical outcome—Dysplasia—Pathology.

NEUROLOGY 2000;54:642–647

Surgical excision of the epileptogenic region is an established therapeutic option for children with medically resistant partial seizures. Although the success of surgical therapy in children is well documented,^{1–8} it is generally assumed that factors associated with good outcome in adults are operative in childhood. Thus, for example, children with normal imaging studies, severe cognitive delay, or extratemporal seizures are denied surgery and referred for alternative treatments including the ketogenic diet and vagal nerve stimulation. Although these alternative therapies may temporarily alleviate or reduce seizures,^{9,10} definitive cure occurs only in a small percentage of individuals.

Several features of pediatric surgical candidates distinguish them from their adult counterparts and are potential determinants of postoperative outcome. For example, the pathologic substrate of chronic childhood epilepsy frequently results from disordered cortical development.^{2,5,7,11,12} Affected patients are prone to early seizure onset and higher incidence of extratemporal seizures.^{3,4,6} An understanding of the influence of pre- and perioperative variables in this setting would facilitate candidate selection and potentially improve the seizure prognosis.

To address these issues, we reviewed our experi-

ence with children younger than 12 years of age undergoing excisional surgery for epilepsy, in order to help determine which pre- and perioperative variables predicted surgical outcome. We analyzed outcome based on age at seizure onset, duration of epilepsy, presence of cognitive impairment, lobe of involvement, presence of a gross structural lesion, histopathologic findings, and completeness of the resection.

Methods. *Patients.* We retrospectively reviewed the presurgical data from a 10-year period in a cohort of 83 patients younger than 12 years of age who underwent focal resection for epilepsy and had follow-up of at least 1 year. All had longstanding medically intractable epilepsy, defined as seizures not responding to adequate trials of maximally tolerated antiepileptic drugs.¹³ We excluded eight patients on the basis of incomplete follow-up ($n = 5$) or death ($n = 3$; two nonsurgical, one postoperative), leaving a total of 75 patients (40 boys, 35 girls) for analysis.

Presurgical evaluation. All patients underwent preoperative video EEG and MRI (0.5-tesla or 1.5-tesla magnet). There were 35 nonlesional and 40 lesional cases. The 1.5-tesla MRI unit was obtained at our institution in mid-1992; 47 cases were performed prior to then (26 lesional, 21 nonlesional), and 28 cases were performed subsequently (14 lesional and 14 nonlesional). For study purposes, focal

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Received December 4, 1998. Accepted in final form September 11, 1999.

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signal abnormalities, gyral anomalies, and hippocampal sclerosis were classified as lesional. Normal studies or non-specific MRI abnormalities such as diffuse atrophy were considered nonlesional. Ictal SPECT was performed in nine patients using standardized protocols.¹⁴ Developmental level was assessed by neurologic evaluation and formal neuropsychological testing.

Fifty-five (73%) patients had extraoperative monitoring. These included all nonlesional cases and 20 (50%) of 40 lesional cases. Resections in the remaining 20 lesional cases were guided by intraoperative electrocorticography. Extraoperative intracranial recording was obtained using subdural electrodes (strips or grids). If appropriate, six or eight contact-depth electrodes were placed strategically into a deep-seated lesion guided by MRI findings or intraoperative ultrasound.¹⁵⁻¹⁷ Functional mapping was performed utilizing direct electrical stimulation whenever epileptogenic activity was recorded in proximity to eloquent cortex.¹⁸

Epileptogenic zone and surgical resections. Resections were planned and carried out on the basis of neuroimaging (CT or MRI) and intracranially recorded electrophysiologic data. Whenever possible, lesions were resected in their entirety, and all cases of preoperatively identified hippocampal sclerosis (n = 6) prompted resections of the mesial temporal structures.

For patients undergoing one-stage excisional procedures, regions of active spiking on the pre-excision electrocorticography were used to define the epileptogenic zone and resection plane. Prominent interictal spiking was considered significant if it showed consistent focality or had rhythmic features, occurred as trains of focal "fast" activity, or was associated with focal attenuation of background. In contrast, infrequent spikes, spikes without consistent focality, or "rim" spikes recorded on the postresection electrocorticograph were ignored.

In implanted patients, the ictal onset zone was the single most critical factor in determining the epileptogenic region, especially in patients with normal imaging studies. We defined ictal onset zones as regions exhibiting focal rhythmic activity, bursts of high-frequency discharges, repetitive spiking, or electrodecremental patterns.¹⁵⁻¹⁷ In addition, secondary foci that consistently activated intraictally¹⁹ (during a seizure) were always included in the resection if they occurred in tissue adjacent or in regional proximity to the primary ictal focus. Sinusoidal theta or delta frequencies, which occurred over widespread regions soon after ictal onset, were not regarded as critical to the completeness of resection. Consistent with the electrocorticograph data, regions of prominent interictal spiking and background abnormalities were also considered significant.

Cortical resections. None of the temporal resections were "standard"; all included the anterior neocortical and mesial limbic structures, tailored posteriorly according to EEG, lesional data, and location of language cortex.²⁰ For seizures that originated posteriorly, resection of the temporal convexity and basal neocortex was extended further posteriorly, with the vein of Labbe being undercut if needed.^{12,20}

Extratemporal resections consisted of complete removal of the lesion combined with corticectomy tailored to the epileptogenic region. Anterior frontal epileptogenic regions were often treated by medial or lateral wedge resections;

posterior frontal, parietal, and occipital foci were more likely to be treated by tailored corticectomy alone.

The resection was judged complete only if the region of significant EEG abnormalities defined previously and the structural lesion (if present) were removed in their entirety. Resections were judged incomplete if either the structural lesion or the electrographically abnormal zone could not be completely excised because of proximity to functionally critical cortex. Completeness versus incompleteness was determined at the time of the resection by the epilepsy team present and was subsequently reviewed by the authors postoperatively using the primary data. In lesional cases, this determination also included the gross appearance of the surgical field at the time of the resection, and the subsequent review of postoperative neuroimaging data.

For study purposes, we analyzed surgical variables according to the following criteria:

- 1) Lobe resected: temporal, extratemporal, or combined multilobar (temporal and extratemporal)
- 2) Staging: one-stage resection performed under electrocorticographic guidance or two-stage surgery utilizing extraoperative interictal and ictal data
- 3) Resection: complete or incomplete
- 4) Histopathology: prenatally (cortical dysplasia, hamartoma, developmental tumor) or peri- or postnatally (infectious, vascular, hypoxic, neoplastic, and hippocampal sclerosis) acquired

Follow-up. Postoperative seizure status was assessed during outpatient visits and telephone contact at least yearly after the first postoperative year. Seizure outcome was classified according to a generally accepted classification scheme⁹: class 1, seizure-free (SF); class 2, greater than 90% seizure reduction; class 3, greater than 50% seizure reduction; and class 4, unchanged (no change or less than 50% seizure reduction). Patients who had persistent seizures after the immediate postoperative period and required multiple manipulations of their antiepileptic drug regimen to gain seizure freedom we classified as class 2. For purposes of data analysis, classes 1 and 2 were classified as good outcomes, and classes 3 and 4 were considered poor outcome. Mean follow-up was 5 years (range 1 to 10 years).

Statistical analysis. Standard χ^2 analysis was performed comparing good versus poor and SF versus non-SF outcome against the variables described. Stepwise, multivariate logistic regression analysis was performed using univariate conditions (presence of cognitive impairment, temporal or extratemporal resection, lesional versus nonlesional resection, developmental versus acquired pathology, complete versus incomplete resection, and good versus poor outcome or SF versus non-SF outcome) and continuous variables (age at epilepsy onset and duration of epilepsy prior to surgery).

Results. Preoperative clinical features. Mean age at seizure onset was 2.8 years (range 0 to 9.7 years), and mean age at surgery was 7.7 years (range 0.5 to 11.9 years). Thirty-two patients (43%) were cognitively impaired. Seizures were symptomatic of a clearly defined cause in 30 patients (40%). Nineteen (25%) had discrete lesions—dysembryoplastic neuroepithelial tumors (DNETs; n = 10), ganglioglioma (n = 3), astrocytoma (n = 3), ependymoma

Table 1 Outcome: comparing type of resection

Outcome	Temporal, n = 29	Extratemporal,* n = 37	Temporal and extratemporal, n = 9	Total, n = 75
Seizure-free	19 (66)	23 (62)	2 (22)	44 (59)
Seizure reduction				
>90%	3 (10)	8 (22)	3 (33)	14 (19)
>50%	3 (10)	1 (3)	2 (22)	6 (8)
Unchanged	4 (14)	5 (14)	2 (22)	11 (15)

Values are n (%).

* Extratemporal: frontal, 29; parietal, 6; occipital, 2.

(n = 1), oligoglioma (n = 1), or sarcoma (n = 1)—six of which were associated with neurocutaneous syndromes: tuberous sclerosis (n = 5) or neurofibromatosis (n = 1). Five patients had peri- or postnatally acquired causes: meningoencephalitis (n = 2), birth asphyxia (n = 1), trauma (n = 1), or stroke (n = 1). Of the six patients with preoperatively identified hippocampal sclerosis, two had additional lesions (hamartomas) in their postoperative pathologic specimens.

Postoperative outcome. Fifty-eight (78%) of the 75 children who underwent resective surgery were improved, and 59% of this group was rendered SF (table 1). The seizure frequencies of only 11 patients (15%) were unchanged after surgery. Approximately one-half of the patients (49%) underwent extratemporal resections, and their outcomes were comparable to patients undergoing temporal lobectomies (table 1): 66% and 62% were SF, respectively. By contrast, combined extratemporal and temporal resections were associated with a less favorable outcome: only two of nine patients (22%) were SF. In terms of seizure freedom, the difference between multilobar and unilobar resections was significant ($\chi^2 = 5.68$, $p = 0.05$).

Outcome of the 40 lesional cases was not statistically different from the 35 nonlesional cases (table 2). Eighty percent of the lesional cases had good outcomes, with 65% becoming SF, compared with 74% and 51%, respectively, for the nonlesional group ($\chi^2 = 0.348$, $p = 0.56$ for good

versus poor; $\chi^2 = 0.82$, $p = 0.36$ for SF outcome). Comparison of outcomes for the lesional cases based on lobe of origin likewise failed to achieve significant differences (table 2).

A much higher proportion of nonlesional resections (19/35) was extratemporal compared with nonlesional temporal resections (9/35). No differences in outcome were found between patients with nonlesional extratemporal resections (89% improved, 63% SF) and nonlesional temporal resections (66% improved, 56% SF) ($\chi^2 = 2.17$, $p = 0.15$ for good versus poor; $\chi^2 = 0.17$, $p = 0.68$ for SF outcome). By comparison, the majority (57%) of nonlesional combined resections had a poor outcome, and only 14% became SF. Differences in outcomes between nonlesional unilobar and multilobar resections approached statistical significance ($\chi^2 = 4.52$, $p = 0.055$ for good versus poor; $\chi^2 = 4.97$, $p = 0.083$ for SF outcome).

The majority (56%) of patients had pathologic features of cortical dysplasia: 79% of this group had a good outcome, and 52% were rendered SF by excisional surgery (table 3). Of six patients with peri- or postnatally acquired pathology, 5 (83%) had a good outcome and 4 (67%) were SF. One patient had no definite pathologic abnormalities in the resected tissue, and one patient had insufficient tissue for analysis.

The outcome of 25 patients with tumors was similar to outcome in other pathologic categories: 19 (76%) had a good outcome and 17 (68%) were SF. There were 19 developmental tumors: DNETs (n = 10), ganglioglioma (n = 3), and hamartoma (n = 6). All patients with DNETs were significantly improved; 7 (70%) became SF. All three patients with gangliogliomas were SF, whereas three (50%) of the six patients with hamartomas were SF. None of the hamartomas had been positively identified as a mass lesion preoperatively. Of the six patients with neoplasms—astrocytoma (n = 3), ependymoma (n = 1), oligodendroglioma (n = 1), and sarcoma (n = 1)—4 (67%) were SF and one had a poor outcome.

There were no statistically significant differences in outcome between patients with cortical dysplasia, tumors, and peri- or postnatally acquired pathology ($\chi^2 = 0.166$, $p = 0.92$ for good versus poor; $\chi^2 = 1.58$, $p = 0.24$ for SF

Table 2 Outcome: comparing type of resection in lesional versus nonlesional cases

Outcome	Temporal, n = 20	Extratemporal, n = 18	Combined, n = 2	Total
Lesional (n = 40)				
Seizure-free	14 (70)	11 (61)	1	26 (65)
Seizure reduction				
>90%	2 (10)	3 (17)	1	6 (15)
>50%	2 (10)	1 (6)	0	3 (8)
Unchanged (%)	2 (10)	3 (17)	0	5 (13)
Nonlesional (n = 35)				
Seizure-free (%)	5 (56)	12 (63)	1 (14)	18 (51)
Seizure reduction				
>90%	1 (11)	5 (26)	2 (29)	8 (23)
>50%	1 (11)	0	2 (29)	3 (9)
Unchanged (%)	2 (22)	2 (11)	2 (29)	6 (17)

Values are n (%).

Table 3 Pathology and outcome

Pathology	Seizure-free	Seizure reduction			Total
		>90%	>50%	Unchanged	
Cortical dysplasia, n (%)	22 (52)	11 (26)	1 (2)	8 (19)	42
Tumors					
Nondevelopmental*	4	1	0	1	6
Developmental†	13	1	4	1	19
Total, n (%)	17 (68)	2 (8)	4 (16)	2 (8)	25
Peri/postnatally acquired, n (%)	4 (67)	1	0	1	6
Normal			1		1
No pathology	1				1

* Three astrocytomas, one ependymoma, one oligoglioma, one sarcoma.

† Ten dysembryoplastic neuroepithelial tumors, six glial-neuronal hamartomas, three gangliogliomas.

outcome), nor was there a statistically significant difference between patients with developmentally based lesions (cortical dysplasia, developmental tumors) compared with peri- or postnatally acquired pathology ($\chi^2 = 1.34$, $p = 0.25$ for good versus poor; $\chi^2 = 0.36$, $p = 0.55$ for SF outcome).

Forty-nine patients were judged to have undergone complete resection of the epileptogenic zone or lesion: 45 (92%) had a good outcome and 37 (76%) were SF. To avoid postoperative functional deficits, 26 (35%) patients had incomplete resections: only 7 (27%) were SF, and 13 (50%) had good outcomes. Thirteen patients (50%) with incomplete resections had poor outcomes, compared with only four patients (8%) with complete resections. One-third of patients with incomplete resections ($n = 8$) had no change in seizure frequency postoperatively. Differences between outcomes in complete and incomplete resection group were highly significant ($\chi^2 = 16.96$, $p < 0.001$ for good versus poor outcome; $\chi^2 = 16.5$, $p < 0.001$ for seizure freedom).

Stepwise, multiple logistic regression analysis. Age at epilepsy onset, duration of epilepsy before surgical resection, presence of cognitive impairment, temporal versus extratemporal resection, lesional versus nonlesional resection, developmental versus acquired pathology, and complete versus incomplete resection were compared with good versus poor outcome. For purposes of analysis, developmental pathology included cortical dysplasia and developmental lesions; acquired pathology included peri- or postnatally acquired pathology and nondevelopmental masses.

On univariate analysis, completeness of resection was the only significant variable. The OR for completeness of resection was 11, indicating that the probability of a poor outcome is 11 times greater after an incomplete resection than a complete resection. Logistic regression for continuous variables, such as duration of epilepsy, did not identify significant predictors of outcome. If completeness of resection was standardized, no other variables were found to be significant. Using seizure freedom versus nonseizure freedom in the analysis, completeness of resection was also the only significant variable, with an OR of 7.8.

Discussion. Our data demonstrate that surgery for pharmacoresistant partial epilepsy is most effective if a complete resection is performed. In contrast

to the adult experience, a temporal lobe location of the seizure focus²¹⁻²⁴ and the presence of a gross structural lesion²¹⁻²⁵ do not significantly improve outcome. Of the pre- and perioperative variables included in our analysis, completeness of the resection was the main determinant of outcome, with 92% of patients who had complete resections achieving good results compared with a 50% improvement rate with incomplete resections.

Multilobar resections had less favorable outcome in terms of seizure freedom, but the significance was not demonstrated in logistic regression analysis. Given the small number of patients in our series with multilobar resections, further analysis of the factors contributing to the significance is limited, but it suggests that widespread areas of epileptogenesis are more difficult to remove in their entirety.

Our results provide support for the adult surgical experience reported by Rossi et al.²⁶ They retrospectively reviewed 138 surgically treated epilepsy patients and found that the presence of a lesion did not predict surgical outcome. Similar to our cohort, complete removal of the structural lesion and epileptogenic zone in adults was highly predictive of success and was independent of the surgical approach. Our findings extend these observations to the pediatric epilepsy surgical population, in which there are high proportions of nonlesional and extratemporal cases, and a higher proportion of candidates with disorders of cortical development.

Our findings also reveal an absence of any correlation between postoperative prognosis and tissue histopathology or lesional status. This finding is unexpected and contrasts with the adult experience.^{22,27-29} This may be, in part, attributable to the greater proportion of dysplastic lesions in our series compared with the higher proportion of adult cases with hippocampal sclerosis, which tend to have good outcomes.^{22,29} Although hippocampal sclerosis is also seen in pediatric surgical candidates with chronic temporal lobe epilepsy,^{2,5,7} it may be encountered in the setting of dual pathology.^{5,7}

Histopathologic lesions are likely to differ in their epileptogenic properties, however, and these differences should influence the surgical approach. For example, anatomically well-defined lesions such as DNETs have been shown to have restricted epileptic zones³⁰ and respond well to one-stage lesionectomy. Without a preoperative pathologic diagnosis, a comprehensive workup with detailed electrophysiologic definition of the epileptogenic region should be obtained.

To perform a complete resection, the electrographically abnormal zone or lesion must be well localized preoperatively. Application of the electrophysiologic definition of completeness is unlikely to be uniform among epilepsy centers. At our center, regions of prominent interictal activity and attenuated background and ictal onset zones, as well as intractably activated regions,¹⁹ are regarded as electrographically significant; remote regions of infrequent spiking are not considered important. Completeness also includes the removal of lesional tissue in its entirety. In this series, lesional status was based primarily on CT or MRI scans. PET data were not included in the preoperative evaluation. In recent years we have incorporated focal ictal SPECT hyperperfusion in our surgical planning, but its role as an outcome predictor was not analyzed.

A high proportion of our population underwent two-stage evaluations with extraoperative subdural recording. This trend is influenced by referral patterns to our institution for more complicated and nonlesional cases. Nearly one-half of our cases were nonlesional, a proportion appreciably higher than the average case mix. Because higher proportions of nonlesional and extratemporal cases are more characteristic of the pediatric population, implantation of a higher proportion of children is likely to continue. Higher field-strength MRI capabilities and the inclusion of ictal SPECT in presurgical protocols should ultimately reduce the number of pediatric patients requiring subdural implantation.

The importance of performing a complete resection has significant implications for surgical planning and patient counseling. If the presurgical evaluation indicates a high likelihood of epileptogenic involvement of critical cortex, our data indicate that the incidence of seizure freedom is markedly reduced. The reduced likelihood of seizure freedom without functional deficit should be discussed openly before implantation, because families may understandably decline surgery. The need for multilobar resection and the potential involvement of functionally critical cortex present the greatest challenges for improving the patient's seizures and require further innovation before similar rates of seizure freedom can be achieved.

Clinical, EEG, imaging, and neuropsychological features assist in the presurgical evaluation for temporal lobectomy in adults,³¹ and multivariate algorithms for predicting success have been described.³² In contrast, no preoperative factors predict seizure

outcome in pediatric temporal lobectomy candidates.³³ Our current findings regarding the importance of a complete surgical resection may help to explain the poor predictive power of preoperative variables. Although the need for a multilobar resection may be a factor, the duration of epilepsy, frequency of seizures, or presence of mental retardation should not influence the determination of surgical candidacy in pediatric epilepsy surgery candidates. Instead, the preoperative evaluation must be individualized, because children with normal imaging studies or cortical dysplasia may be strong surgical candidates.

Acknowledgment

The authors thank Melissa Scheiderer for manuscript preparation and Dr. John Hayes for statistical analysis.

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Reduction of corpus callosum growth after severe traumatic brain injury in children

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Article abstract—*Objective:* To study effects of closed head injury (CHI) severity on development of corpus callosum (CC) in children, using MRI. *Background:* Vulnerability of CC to diffuse axonal injury has been shown in adults and children by neuropathologic and MRI studies. Given continued development of CC through the second decade, serial MRI could characterize effects of CHI on CC growth in children. *Methods:* MRI performed at 3 and 36 months after severe (mean age = 10.3 years, n = 25) and mild to moderate (mean age = 9.7 years, n = 28) CHI. Mild to moderate and severe CHI groups did not differ in demographic features. Morphometry of T1-weighted midsagittal CC by two operators with satisfactory interrater reliability yielded uncorrected and corrected CC volume. *Results:* An interaction of occasion with CHI severity was present as CC area decreased from 3 to 36 months in severely injured children and increased in the mild to moderate CHI group. Uncorrected CC area was correlated with acute CHI severity and functional outcome at 36 months postinjury. *Conclusions:* Morphometric measurement of CC area provides a useful index of diffuse injury, which is related to functional outcome of CHI in children. **Key words:** Corpus callosum—Traumatic brain injury—Children.

NEUROLOGY 2000;54:647–653

Corpus callosum (CC) hemorrhages and shear strains are characteristic features of diffuse axonal injury (DAI) associated with closed head injury (CHI). These features are especially apparent in association with high-velocity traumatic forces such as

motor vehicle accidents and pedestrian–motor vehicle injuries that impart rotational acceleration, produce shear strain, and result in immediate, severe impairment of consciousness.^{1–5} Neuropathologic³ and MRI^{6,7} studies have documented that CC lesions

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Supported in part by NIH grant NS21889, “Neurobehavioral Outcome of Head Injury in Children,” and M01-RR02558, the General Clinical Research Center, Baylor College of Medicine, and the University Clinical Research Center at Hermann Hospital.

Received May 17, 1999. Accepted in final form September 3, 1999.

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Table Demographic and clinical features of children with head injuries

Patient characteristics	Mild to moderate, n = 28	Severe, n = 25	Statistics
Age at injury, y, mean $\pm$ SD	9.7 (2.7)	10.3 (3.1)	F(1,51) = 0.6, p = 0.44
Follow-up interval, mo, mean $\pm$ SD	33.6 (2.8)	33.5 (1.5)	F(1,51) = 0.1, p = 0.83
Parent education, y, mean $\pm$ SD	14.4 (2.4)	14.0 (3.4)	F(1,50) = 0.3, p = 0.59
Glasgow Coma Scale score, mean $\pm$ SD	13.5 (1.7)	5.7 (2.0)	F(1,51) = 241.1, p < 0.01
Sex, %			
Male/female	71.4/28.6	60.0/40.0	$\chi^2(1)$ = 0.8, p = 0.38
Mechanism of injury, %			
High velocity	35.7	72.0	$\chi^2(1)$ = 6.9, p = 0.01
Low velocity	64.3	28.0	
Classification of head injury according to CT findings ¹⁶			
Diffuse injury I, n (%)	20 (71.4)	5 (20.0)	
Diffuse injury II, n (%)	7 (25.0)	12 (48.0)	$\chi^2(3)$ = 3.9, p = 0.04
Diffuse injury III, n (%)	0	4 (16.0)	
Diffuse injury IV, n (%)	0	0	
Evacuated mass, n (%)	1 (3.6)	4 (16.0)	
Nonevacuated mass	0	0	

are present in approximately 15% to 20% of adults and children sustaining severe CHI. Consistent with the shear strain gradient of DAI⁸ and the presence of intracranial dural compartments (e.g., falx, tentorium cerebri), focal CC lesions occur most frequently in the posterior body and splenium.⁷ In addition to focal CC lesions, CC atrophy in adults after severe CHI has also been reported in MRI^{9,10} and neuropathologic¹¹ studies.

With MRI evidence for continued development of the posterior CC through at least the second decade¹²⁻¹⁴ and the rapid phase of myelination during years 1 through 10, we postulated that severe CHI in young children would disrupt CC development. Apart from potential use as a marker of residual brain injury after CHI, deficient CC development could compromise performance of activities that stress interhemispheric integration. To investigate the effects of CHI severity on development of the CC, we obtained repeat MRIs at 3 and 36 months postinjury in children and adolescents who had sustained severe or mild to moderate CHI. Second, we studied the relationship between postinjury growth of the CC and functional outcome of CHI.

Patients and methods. *Patients.* Fifty-three pediatric CHI patients were recruited from consecutive admissions to the neurosurgical services at Parkland/Children's Medical Centers in Dallas and Hermann Hospital in Houston (table). Selection criteria included age 5 to 15 years at the time of injury, age 5 to 18 years at the time of initial and follow-up MRIs, and hospitalization for CHI with CT within 24 hours of injury. This study was part of a larger project that stipulated a lower age limit of 5 years because it was not feasible to administer the cognitive tests to younger children. Exclusion criteria were previously diag-

nosed neuropsychiatric disorder, including previous hospitalization for CHI, epilepsy, or developmental disorders such as attention deficit-hyperactivity disorder. Severe CHI was defined by a postresuscitation Glasgow Coma Scale (GCS)¹⁵ score of 8 or less without a previous interval of normal consciousness irrespective of CT findings, whereas a mild CHI was defined by lowest GCS score of 13 to 15, normal neurologic examination, normal CT findings other than a linear skull fracture, and no extracranial injury requiring general anesthesia for surgical repair. Moderate CHI was defined by a lowest postresuscitation GCS score of 9 to 12 irrespective of other findings. In view of neuropathologic and MRI studies^{10,11} in adults indicating that CC atrophy occurs primarily in CHI producing immediate onset of coma, we compared the findings of the severely injured patients (n = 25) with the results obtained from the combined mild to moderate CHI groups (n = 28). All 53 head-injured children enrolled in a large project on outcome of pediatric CHI who met the eligibility criteria, including initial and follow-up MRI, were included in this study. Consequently, no power analysis was performed.

We performed analysis of variance (ANOVA) tests to examine the differences in demographic features between the severe and mild to moderate CHI groups. As reflected in the table, there were no significant differences in demographic features. In patients who had more than one CT scan, the first CT scan was selected for classification of injury, with the exception of those patients with delayed lesions, in whom the findings from later CT scans were used. Classification of CHI¹⁶ according to the CT findings obtained during the initial hospitalization disclosed that diffuse injury associated with compressed cisterns as well as small focal lesions with a midline shift of 0 to 5 mm characterized approximately two thirds of the severely injured patients, compared with only one fourth of the mild to moderate CHI group (see table). In contrast, approximately 71% of the mild to moderate CHI group and 20% of

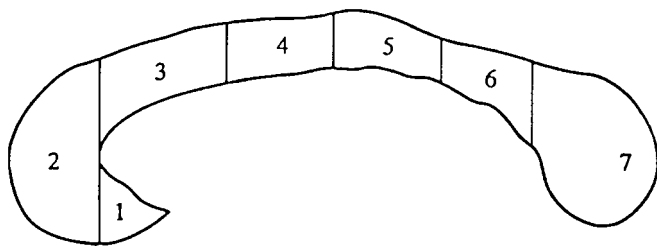


Figure 1. Corpus callosum (CC) sections corresponding to the divisions used by Witelson²¹ in a postmortem morphologic study of the CC.

the severely injured patients had no visible pathologic condition on CT. The distributions of injury type were significantly different for the mild to moderate and severe CHI groups (see table). Neuropathologic evidence¹¹ suggests that the velocity of the traumatic forces is a major determinant of DAI severity. Consequently, we classified the external causes of injury as high-velocity (i.e., motor vehicle crash, pedestrian-vehicular injury) or low-velocity (i.e., falls from a short distance, recreational injury). High-velocity accidents produced nearly two-thirds of the severe head injuries as compared with approximately one third of the mild to moderate CHIs. There were four patients with evacuated epidural hematomas, including three severely injured children and one moderate CHI patient. One severely injured patient had an acute subdural hematoma evacuated.

Although we acknowledge that confirmation of DAI requires neuropathologic examination, recent histologic investigation¹⁷ has documented white matter injury in adults sustaining mild traumatic brain injury. With evidence for a continuum of DAI severity, we postulated that children who were rendered comatose immediately after injury generally sustained more severe DAI than children who had mild, brief impairment of consciousness.

Assessment of functional outcome. Functional outcome of CHI was assessed at 36 months postinjury using the Vineland Adaptive Behavior Scale.¹⁸ National percentile scores were analyzed for the Adaptive Behavior Composite, Communication, and Daily Living domains of the Vineland. With the widespread use of the Glasgow Outcome Scale (GOS)¹⁹ to evaluate global recovery from CHI, we also analyzed the GOS category in relation to total CC area.

Magnetic resonance imaging. MRI for all 53 patients in this study was scheduled to approximate 3 months and

36 months postinjury (see table). These MRI intervals corresponded to scheduled follow-up assessments, thus limiting the number of return visits requested of the families. Based on previous MRI investigation of pediatric CHI,⁶ 3 months was selected for the initial MRI because transient effects such as edema resolve by this time, and the child is better able to cooperate with the scan. Although neurodegenerative changes can be seen by 3 months after severe CHI,⁵ the 3-year follow-up was selected to investigate the impact of severe CHI on CC development. All CC measurements were obtained from digitized midsagittal T1-weighted images. Technologic advances in MRI permitted a reduction in slice thickness from 5 mm to 1.5 mm over the 5-year course of study (1990 to 1995).

Anatomic measures. A single T1 midsagittal slice in which the fourth ventricle, pituitary stalk, lower brainstem, and massa intermedia were visible was used to measure the CC area and estimate the total brain area. Two operators who had no information regarding the severity of acute traumatic brain injury performed the morphometry on a Sun Sparc Classic computer (Sun Microsystems, Mountain View, CA) using a program written in the Sunview windows environment (Sun Microsystems) with mouse-activated controls for selecting the various steps of each analytical sequence.²⁰ After manually outlining the CC, a semiautomated program determined the anteriormost and posteriormost points by analyzing the x-coordinates of the CC outline. The program then drew a line connecting the anteriormost and posteriormost points (the AP line). The CC was then divided into seven sections²¹ (figure 1). The first two sections were determined by the "hook" forming the rostrum and genu regions; this dividing line was perpendicular to the AP line. The other regions were determined by perpendicular lines drawn from the AP line. These dividing lines were drawn at 20%, 33%, 50%, 67%, and 80% from the center of the total AP line length. Areas were calculated in number of pixels from the x-y coordinates of the points of the outline of each region, i.e., polygon.^{22,23} CC lesions were outlined manually with a computer mouse and the software program computed the lesion area in number of pixels. Lesion areas were subtracted from the callosal area containing the lesion. Total and regional CC areas were converted to mm² by multiplying the number of pixels by pixel size. An estimate of brain area was obtained by manually outlining the brain on the midsagittal image. The software program computed the brain area in pixels. To correct for individual

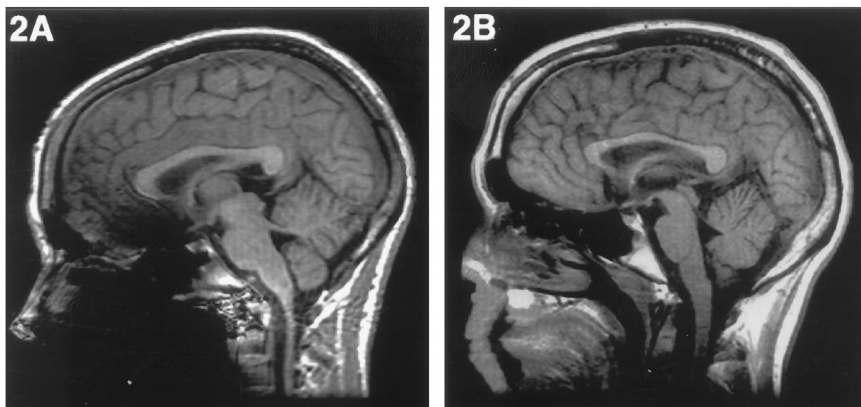


Figure 2. Reduced corpus callosum (CC) area after severe head injury. A 14-year-old boy was struck by a car while riding a go-cart. Lowest postresuscitation Glasgow Coma Scale score was 6. CT scan on day of injury showed punctate areas of high density in both prefrontal regions and in left parietal area, but no mass lesion. MRIs at 3 months (A) and 36 months (B) postinjury were consistent with diffuse brain injury and a clear reduction in CC area.

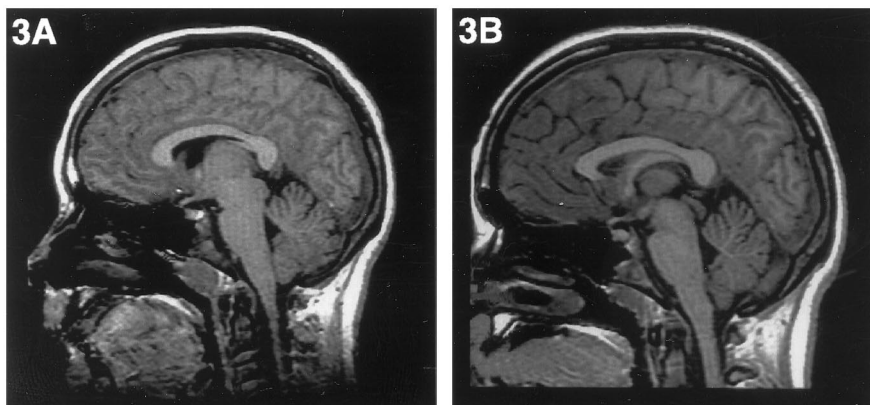


Figure 3. Minimally changed corpus callosum (CC) after mild to moderate head injury. A 13-year-old girl fell and had a Glasgow Coma Scale score of 15 on admission to hospital. CT showed fractures of petrous and posterior parietal bones. MRI at 3 months (A) showed a small area of pneumocephalus in right parietal region with prominence of right frontal and left temporal horns. MRI at 36 months (B) showed minimal change in area of CC and evidence of bifrontal atrophy in the gyrus rectus and inferior frontal gyrus.

differences in brain size, a ratio ($[\text{CC area}]/[\text{brain area}] \times 100$) was computed.

Interoperator reliability. One operator performed the primary measurements, and as a reliability check, a second operator performed measurements on eight children. Intraclass correlation coefficients ranged from 0.76 (rostrum) to 0.99 (splenium), all within acceptable levels.

Statistical analysis. CC segment areas measured from the 3- and 36-month MRIs were analyzed using a repeated measures design. Interactions among CHI severity group (mild to moderate, severe), occasion (3, 36 months), and region (CC1–CC7) were examined, as were the main effects of severity group, occasion, and region. Analysis was performed for the area measurements and for the ratio of the area measurement to total brain volume on the MRI slice depicting the CC. Because age at injury was not predictive of CC areas, we excluded age from the final models fitting CC areas. Pearson correlations were computed to assess the interrelationships among the CC measures and functional outcome at 36 months postinjury.

Results. Figures 2 and 3 show the midsagittal MRI at 3 and 36 months of representative patients in the severe CHI and mild to moderate CHI groups. The images in figure 2 show a reduction in CC area on the MRI obtained at 36 months postinjury as compared with the findings at 3

months after a severe diffuse brain injury in an adolescent. In comparison, figure 3 shows minimal change in CC area between 3 and 36 months in an adolescent whose GCS score reflected normal consciousness at the time of hospital admission.

Uncorrected morphologic data. Analysis of CC area data disclosed no triple interaction of occasion (3, 36 months), group (mild to moderate, severe CHI), and region (CC1–CC7), $F(6,663) = 1.1$, $p = 0.39$. However, an occasion $\times$ group interaction was present, $F(1,663) = 11.1$, $p = 0.001$, indicating that changes from 3 to 36 months postinjury varied according to CHI severity. Figure 4 shows an increase in total CC area from 3 to 36 months postinjury in the mild to moderate CHI group paralleled by a reduction in total CC area in the severely injured patients over the same interval.

A group $\times$ region interaction was also present, $F(6,663) = 9.9$, $p < 0.0001$, indicating that the effects of CHI severity varied with the region of the CC. Figure 5 plots the mean areas of each CC region for the mild to moderate and severe CHI groups collapsed across the two MRIs. p Values of group comparisons within each CC segment are given in figure 5. Although the anteroposterior gradient of differences in regional area between the mild to moderate and severe groups was nonlinear, the significant differences in regional area were all posterior to the genu.

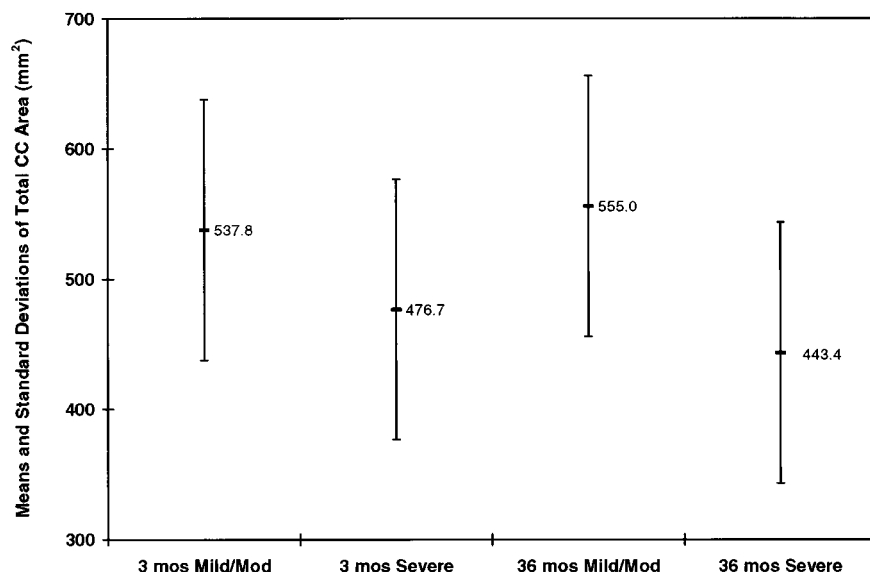


Figure 4. Mean and standard deviation of total corpus callosum (CC) area obtained on MRI at 3 months and 36 months postinjury and plotted by severity of head injury group. Interaction of group with occasion is reflected by the reduction in total CC after severe head injury in contrast to the increased CC area over time after mild to moderate head injury.

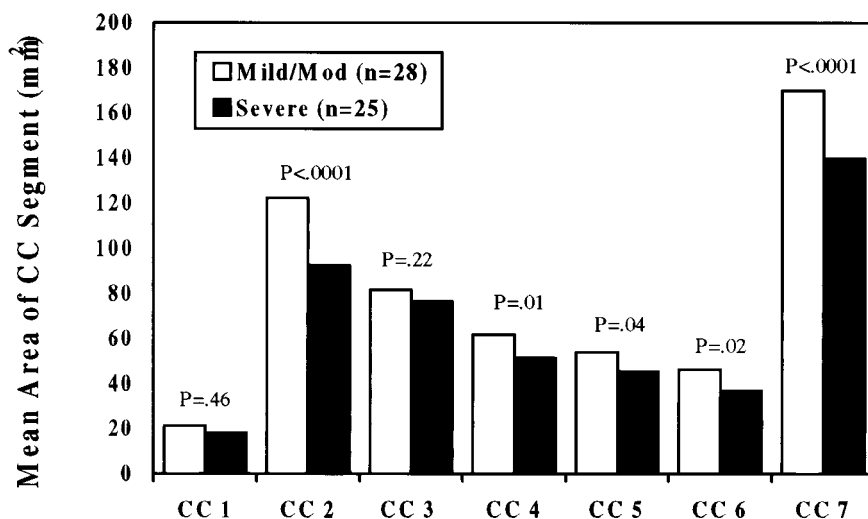


Figure 5. Mean area plotted for seven segments of corpus callosum (CC) according to severity of head injury. Data are collapsed across the 3-month and 36-month MRI scans because there was no interaction of occasion with segment, nor was there a triple interaction of occasion, segment, and group. An interaction of severity group with segment is reflected by variation across the segments in the magnitude of group differences in area. Pairwise group differences in area are denoted by p value. The seven sections correspond to the divisions used by Witelson²¹ in a postmortem morphologic study of the CC.

Means and standard deviations of the changes in total CC area between the 3- and 36-month MRIs were calculated for mild to moderate versus severe patients. The mild to moderate patients had an 18.2 mm² (SD = 40.6) increase in CC total area, and the severe CHI group had a 33.3 mm² (SD = 70.4) decrease in total CC area. An ANOVA model showed a difference of change in total CC area between the CHI severity groups, $F(1,51) = 10.3$, $p = 0.002$. The difference in whole brain size between the 3- and 36-month MRIs was fitted into a regression model. In contrast to the results from CC data, only the effect of age at injury was found, $F(1,49) = 6.8$, $p = 0.01$, disclosing a larger increase of whole brain size among younger CHI patients as compared with older head-injured children.

Corrected morphologic data. The same analysis was performed on the CC area data adjusted for brain volume to control for differences in brain size. Similar to the unadjusted data, there was an occasion $\times$ group interaction, $F(1,663) = 8.2$, $p = 0.004$, as well as a group $\times$ region interaction, $F(6,663) = 8.3$, $p < 0.0001$.

Relationship of corpus callosum area at 36 months postinjury to acute CHI severity and outcome. *Glasgow Coma Scale and Vineland Adaptive Behavior Scale.* A correlation matrix was obtained for total CC area on the 36-month MRI, age at injury, GCS score, and the 36-month Vineland Adaptive Behavior Scale¹⁸ to provide an index of functional outcome. Severity of impaired consciousness accounted for approximately 11% of the variance in total CC ($r = 0.34$, $p = 0.01$). Total CC area was also correlated with functional outcome as measured by national percentile scores on the Adaptive Behavior ($r = 0.42$, $p = 0.003$) and Communication ($r = 0.37$, $p = 0.008$) domains of the Vineland Scale. However, neither the Vineland Daily Living Skill ($r = 0.15$, $p = 0.29$) percentile score nor age at injury ($r = 0.15$, $p = 0.26$) had a consistent relationship with the CC measures.

Glasgow Outcome Scale. To analyze the relationship of CC morphology at 36 months postinjury and functional outcome, the distribution of GOS¹⁹ category at 3 years postinjury, which was limited to good recovery, moderate disability, and severe disability, was dichotomized as good recovery versus disability (moderate or severe). There was no significant difference in age at follow-up between the

children who had a good recovery (mean age = 13.4 years, SD = 2.8) and patients who were disabled (mean age = 12.4 years, SD = 3.0), $F(1,50) = 1.2$, $p = 0.29$. The association between GOS and CC segment was examined by using a logistic regression model with GOS category as dependent variable and total CC area, age at injury, and GCS as independent variables. The analysis disclosed a significant main effect of CC, $\chi^2(1) = 5.6$, $p = 0.02$, indicating that a larger total CC area is predictive of a good recovery after adjusting for age and GCS effects.

Discussion. Neuropathologic studies of CHI in adults and children^{11,24} and experimental nonimpact head injury models in nonhuman primates² have reported that focal hemorrhages of the CC are characteristic features of DAI. MRI performed at least 3 months postinjury^{6,7} has confirmed that CC lesions are present in approximately 20% of children sustaining severe CHI. The posterior body and splenium are the most frequent sites of lesion, a pattern that has been attributed to the distribution of shear strain produced by rotational acceleration and the effects of intracranial dural compartments such as the falx.^{4,8} Based on recent MRI findings indicating that development of the CC in humans continues through adolescence and the middle 20s,^{12,13} we postulated that severe CHI in children would disrupt its growth. Repeat MRI 3 years after severe CHI showed a mean reduction in area of the CC. Our finding that CHI severity interacted with occasion reflects divergent changes in CC size in children who sustained severe CHI relative to patients of similar age who had injuries that did not produce coma as defined by the GCS. Consistent with this pattern, we¹⁰ found that MRI evidence of CC atrophy in adults was present approximately 4 months after severe, but not mild to moderate, CHI. The relationship of CC area at 3 years postinjury to the lowest postresuscitation GCS score provides indirect support for the role of DAI. High-velocity traumatic forces (e.g., motor vehicle crashes), which predomi-

nated in the severe CHI group, could have imparted rotational acceleration to the brain, thus producing shearing of the interhemispheric fibers.

Consistent with developmental MRI studies based on longitudinal and cross-sectional data,^{12,13} we found that children exhibited growth of the CC over the 3 years after a mild to moderate CHI as defined by their GCS score. Although caution is advised in comparing morphometric MRI data across studies because of variation in measurement methods, CC growth in the mild to moderate CHI group approximates the findings reported for healthy children.^{12,13} With rapid myelination during the first decade of life,²⁵ it is conceivable that severe CHI occurring in the birth to 5 years age range could produce greater disruption of CC growth than comparable injuries in older children or adolescents. Although we found no evidence for an age effect or interaction of age with CHI severity to support this possibility, our selection of children who were at least 5 years old at the time of the initial MRI could have mitigated age effects. Consistent with this interpretation of our data, neuropathologic studies of CHI in children²⁴ and adults¹ have reported that macroscopic CC lesions occur with severe grades of DAI that are typically produced by high-velocity traumatic forces such as motor vehicle crashes.¹ Similarly, findings showed² in a subhuman primate model using nonimpact rotational acceleration that severe DAI associated with coma was necessary to produce CC lesions. Although a spectrum of DAI severity has been documented that included a lucid interval in one third of DAI patients, macroscopic CC lesions were confined to the most severe injuries. In contrast to the relatively brief durations of survival of patients reported in these neuropathologic studies, the 3-year post-injury interval before performing the second MRI provided a longer period for degenerative changes to progress. Consequently, we postulate that prolonged degenerative changes contributed to the lack of growth and even shrinkage of CC in the most severely injured children.

Because the CC interconnects all major subdivisions of the cerebral cortex including fibers from associative regions,²⁶ its growth could facilitate maturation of integrative skills. Although we found that the CC area at 3 years postinjury was related to functional outcome of CHI, we cannot impute causation from the morphometric-outcome correlations.

Technical limitations of this study include partial volume effects¹³ in measuring the CC area. Because this source of error was common to both the severe and mild to moderate CHI groups, it is unlikely to explain our findings. Despite semiautomated measurement of the CC segments, interobserver variation in drawing the line connecting the anterior and posterior commissures could have occurred. However, we found satisfactory agreement between the measures of CC area obtained from two observers.

From a clinical perspective, we suggest that CC morphometry could provide a reliable index of cere-

bral white matter injury in children. It is also plausible that morphometric analysis of the CC would be sensitive to pharmacologic interventions designed to preserve dysfunctional axons and mitigate secondary brain injury.

Acknowledgment

The authors thank Jackie Duron, Sharon Robinson, and Gilda Warren for assistance with manuscript preparation.

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Hypersomnia after head–neck trauma

A medicolegal dilemma

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Article abstract—*Objectives:* To evaluate the severity of daytime sleepiness in patients with a history of head trauma who complain of daytime somnolence, to investigate polygraphic abnormalities during nocturnal sleep, and to determine whether daytime sleepiness was the cause or consequence of the head trauma. *Methods:* The authors performed a systematic evaluation of 184 patients comprised of clinical interviews, sleep disorders questionnaires, sleepiness and depression scales, medical and neurologic evaluations, sleep logs with actigraphy, nocturnal polysomnography, and the Multiple Sleep Latency Test (MSLT). Assessments of sleepiness before the accident were based on bed partner interviews, coworker and employer reports, health reports, driving records, and employment history that included absenteeism. *Results:* Post-traumatic complaint of somnolence was associated with variable degrees of impaired daytime functioning in more than 98% of patients. Patients who were in a coma for 24 hours, who had a head fracture, or who had immediate neurosurgical interventions were likely to have scores > 16 points on the Epworth Sleepiness Scale (ESS) and ≤ 5 minutes on the MSLT. Pain at night was an important factor in nocturnal sleep disruption and daytime sleepiness. Sleep-disordered breathing was a common finding and was the only finding in whiplash patients with daytime sleepiness. Extensive evaluation of pretrauma behavior supported the conclusion that the onset of symptomatic sleep-disordered breathing was associated with the trauma. The patients who showed a “compulsive presleep behavior” were severely impaired in performing their daily activities. *Conclusions:* A systematic approach is required when dealing with patients complaining of hypersomnia following a head–neck trauma. **Key words:** Daytime somnolence—Legal problems—Head–neck trauma—Sleep-disordered breathing—Apathetic syndrome.

NEUROLOGY 2000;54:653–659

Post-traumatic hypersomnia is a difficult medicolegal challenge and poses diagnostic and long-term prognosis questions. The first patient of “idiopathic and traumatic hypersomnia” was reported in 1941.¹ Fifteen years ago, we reported 20 patients with “post-traumatic hypersomnia.”² Because of this report, we saw an increase in referrals of patients complaining of hypersomnia after a head–neck trauma. We need to assess objectively whether the patient is truly sleepy and to determine whether there is any relationship between the head–neck trauma and any currently observed daytime sleepiness. Review of the literature offered few guidelines for responding to these issues. This patient series outlines our strategic approach to evaluate 184 adult patients during a 15-year period.

Methods. *Study population.* The study population consists of 184 adults: 48 women with a mean age of 34 ± 8

years (range, 18 to 52 years), and 136 men with a mean age of 39 ± 12.5 years (range, 18 to 54 years). A total of 103 patients had insurance, financial, and medicolegal issues linked with their head trauma and their subsequent report of excessive daytime sleepiness. The mean time between the head trauma and the referral to the sleep clinic was 15.5 ± 5 months.

The patient population was divided into subgroups based on the reported trauma and duration of loss of consciousness documented in the initial medical charts. The population is presented in table 1. The type of trauma was related to vehicular accident (car, truck, bus, motorcycle, boat, and plane) in 124 patients, falls (scaffold, ladder, tree, and horseback riding) in 50 patients, and inflicted violence in 8 patients. For two patients, the cause of trauma was unknown at the time of review.

When seen at our clinic, 13 patients presented with palsy (involving arms, legs, or both), 9 patients reported daily headaches, 11 patients had a controlled seizure disorder, and 2 patients reported residual cervical pain after

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Supported by an academic award from the Center for Sleep Research of the National Institutes of Health (C.G.).

Received March 22, 1999. Accepted in final form September 25, 1999.

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Table 1 Type of presenting trauma in the study population

Nature of trauma	Total, n (%)	Men, n (%)	Women, n (%)
Coma > 90 min, usually $\geq$ 24 hr	70 (38)	54 (77)	16 (23)
Head fracture	28		
No head fracture	42		
Loss of consciousness < 90 min	90 (49)	67 (74)	23 (26)
Head fracture	23		
No head fracture	67		
Neck trauma	16 (9)	11 (69)	5 (31)
Head fracture	0		
No head fracture	16		
Grogginess	6 (3)	4 (67)	2 (33)
Head fracture	0		
No head fracture	6		
Unknown	2 (1)	0 (0)	2 (100)
Total	184 (100)	136 (100)	48 (100)

cervical vertebrae fusion (C5 through C7). Forty-eight patients were receiving antiepileptic medications (most commonly phenytoin), and five patients had been prescribed a specific serotonin reuptake inhibitor.

Evaluation. All patients underwent an evaluation that included the following: clinical interview, bed partner interview, neurologic and medical examination (with body mass index [BMI]), the Sleep Disorders Questionnaire, the Stanford Sleepiness Scale, the Epworth Sleepiness Scale (ESS), sleep-deprived awake and asleep EEG, 1 week of sleep logs, nocturnal polysomnography, and the Multiple Sleep Latency Test (MLST). The clinical evaluation included a review of the initial medical chart, progress notes, treatments, reeducation schedules, EEGs, and MR images and CT scans acquired before referral to our clinic. The following tests were added as needed: the Hamilton Depression Scale, the Beck Scale, the Wilkinson Addition Test, the Tapping Test (reaction time), the Psychovigilance Test (reaction time), brain/cervical MRI, cephalometric radiographs, human leukocyte antigen (HLA) typing, ambulatory polysomnographic monitoring, and actigraphy. These tests were performed only on subgroups of the population. Nocturnal polysomnography was always performed for 8 nocturnal hours based on the patients' bedtime.

Analysis. Polysomnographs were scored according to international criteria: sleep/wake was scored using the Rechtschaffen and Kales international atlas,³ sleep-disordered breathing and apneas were scored using Guilleminault's classification,⁴ and the different scales were scored according to the recommended scoring criteria. The MLST was performed and scored according to the criteria established by Carskadon and Dement.⁵

Statistical analysis. Analysis of variance and multiple logistic regression (stepwise regression) were performed on group data using the SPSS statistical package (Chicago, IL). Cluster analysis was performed on specific variables. A two-tail significance level was used.

Table 2 Patient-reported clinical symptoms

Symptom	n (%)
Impaired daytime functioning	181 (98.4)
Difficulty concentrating	159 (86.4)
Hypnagogic hallucination	98 (53.3)
Restless sleep	91 (49.4)
Disturbed nocturnal sleep	90 (48.9)
Depressive affects	79 (43.0)
Nightmares	76 (41.3)
Regular loud and heavy snoring	66 (35.9)
Nocturnal dry mouth	61 (3.2)
Mouth breathing	60 (32.6)
Nocturnal drooling	56 (30.4)
Recurring nocturnal night sweats	49 (26.6)
Headaches	
At least one per week	39 (21.2)
Daily	9 (4.9)
Nocturnal pain	36 (19.6)
Leg and body jerks during sleep	34 (18.5)
Night terrors	31 (16.8)
Sleep paralysis	27 (14.7)
Abnormal behavior during sleep (REM behavior)	16 (8.7)
Seizure during sleep	11 (6.0)

Results. *Employment records.* Twenty-six months after trauma, 91 patients (49.5%) were unemployed and had been unable to resume normal activities; 39 patients (21.2%) maintained employment but were below the level held before the accident, resulting in a loss of income; and 54 patients (29.3%) had full-time employment that was similar to their employment before the accident. Twenty-two of these 54 patients had neck injuries or reported head trauma without loss of consciousness (2 patients had an unknown status), and 20 more reported a brief loss of consciousness that had passed by the time they were seen in the emergency room.

Clinical symptoms. The symptoms indicated by the patients during the clinical interview and in the sleep disorders questionnaires are outlined in table 2 (one patient might have more than one symptom). These symptoms were grouped into the following categories: difficulties with daytime functioning (n = 181, 98.4%), disturbed nocturnal sleep (49%); two or more of the following symptoms: disturbed nocturnal sleep, hypnagogic hallucinations, sleep paralysis, and abnormal REM sleep behavior), depressive effects (43%; including nightmares and night terrors), abnormal breathing during sleep (33%; including snoring, dry mouth, mouth breathing, and drooling), seizure during sleep (6%; usually intermittent), and pain during the night (19.6%).

All 184 patients had daytime sleepiness, with 181 patients emphasizing that the sleepiness limited their ability to perform at work (31 of these patients had sleepiness that reduced all of their daytime activities drastically). Thirteen patients showed clinically significant weakness secondary to the trauma (mono-, hemi-, or tetraparesis).

Table 3 Relationship between Multiple Sleep Latency Test (MSLT) and Epworth Sleepiness Scale (ESS)

ESS (n = 174)	MSLT (n = 184)			
	≤5 min (n = 51)	>5 min and <10 min (n = 101)	≥10 min and ≤12 min (n = 24)	>12 min (n = 8)
≥16 (n = 61)	48	12	0	1
≥10 and <16 (n = 102)	3	68	24	7
<10 (n = 11)	0	11	0	0
No ESS	0	10	0	0

Note that the MSLT was developed and tested before the creation of the ESS—a subjective scale of sleepiness. This explains why several patients have no ESS score. Normal ESS score, <10; normal MSLT, ≥10 minutes.

Family history. Family history was negative for narcolepsy, but was positive for restless leg syndrome in one patient, and was positive for a history of treated sleep-disordered breathing in another.

Evaluation of nocturnal sleep. There were 184 nocturnal polygraphic recordings and 180 sleep logs obtained for at least 1 week, with concomitant actigraphy in 104 patients.

Polysomnography had a mean total dark time of 480 ± 11 minutes. During the week of sleep logs, the mean total bedtime (from first lights out to last awakening) was 592 ± 71 minutes. Based on the logs, 39 patients (27.7%) slept less than 360 minutes (313 ± 41 minutes), 62 patients (34.4%) slept between 360 and 480 minutes (433 ± 18 minutes), and 79 patients (43.9%) slept 480 minutes or more (548 ± 38 minutes). Actigraphy was related closely to sleep log results only for the group reporting 360 minutes of sleep time (mean inactivity time, 325 ± 52 minutes). These patients attributed their nocturnal sleep disruption to pain, inability to turn in bed easily, nocturia, headaches, and (rarely) seizures.

The comparison between sleep logs and actigraphy in the other groups indicated that, depending on the night, patients had discrepancies between movement time measured with actigraphy and estimated total sleep time. There was no systematic error, but these patients presented at least a one-night discrepancy of 40 minutes or more, whereas the patients who slept 360 minutes had a maximum discrepancy of 20 minutes on the surveyed nights.

Evaluation of sleepiness. MSLT scores were divided into four groups and ESS scores⁶ were divided into three groups (table 3). We performed a cluster analysis with ESS and MSLT scores as variables. We found that the only significant cluster involved patients with an MSLT of 7 minutes and an ESS score of 16. All patients (n = 60) with an MSLT of 7 minutes had a score of 16 or higher on the ESS, and all but one patient with an ESS score of 16 had an MSLT score of 7 minutes. A comparison of MSLT and ESS scores by subgroup is presented in table 4. Some

Table 4 Subgroup analysis

Subgroup	ESS score	MSLT, min
Head fracture (n = 51)	19.3 ± 4	5.5 ± 2.3
Coma > 24 h (n = 42)	16.8 ± 4.2	6.7 ± 5.5
Sleep-disordered breathing	12.8 ± 2.1	7.9 ± 2.2

ESS = Epworth Sleepiness Scale; MSLT = Multiple Sleep Latency Test.

patients with a clinical complaint of daytime sleepiness had MSLT scores of more than 10 minutes. We subdivided these patients into two groups considering that an MSLT score of 12 minutes on a single MSLT did not fit the usual polygraphic criteria for excessive daytime sleepiness.

Sleep-disordered breathing as a cause of sleepiness. Based on the combined findings of the clinical evaluation, the polygraphic evaluation, and a reduction of daytime sleepiness (as measured by MSLT) after treatment with nasal continuous positive airway pressure, 59 patients were diagnosed with sleep-disordered breathing. Of these patients, 50 had an apnea-hypopnea index (AHI) of 31 ± 11 events per hour of sleep. More than 94% of the events were of the obstructive or mixed types. The mean BMI of the patients was 26.7 ± 1.8 kg/m².

Nine patients were diagnosed with sleep-disordered breathing despite an AHI of less than five events per hour of sleep. These patients had abnormal increases in respiratory effort as measured by esophageal pressure (P_{es}) recordings, defined by a maximum peak end inspiratory P_{es} more than 2 SDs during nonobstructed breathing. The maximum P_{es} monitored was 61 cm H₂O (baseline, 6 cm H₂O). The mean maximum P_{es} was 34 ± 7 cm H₂O followed by an alpha EEG arousal. This pattern was noted in at least 10% of the recordings in each of the nine patients. After the arousal, the respiratory effort decreased (P_{es} reversal). These patients were identified as having upper airway resistance syndrome.^{7,8}

Table 5 looks at sleep-disordered breathing in subgroups of the total population. Forty-nine of the 59 sleep-disordered breathing patients had maxillomandibular and orofacial features considered to be risk factors for abnormal breathing during sleep. (It must be emphasized that these anatomic features are only considered “risk factors” and some of them, as indicated later, were also found in subjects without sleep-disordered breathing.) These fea-

Table 5 Relationship between positive diagnosis of sleep disordered breathing (SDB), initial trauma, and craniofacial anatomic risk factors

Characteristic	n	Positive diagnosis of SDB, n (%)
Clinical symptoms of SDB	66	59 (89.4)
Craniofacial “risk factors”	69	49 (83)
Coma without head fracture	42	7 (16.7)
Loss of consciousness < 90 min	90	36 (40)
Whiplash	16	16 (100)

tures include a long face ($n = 63$), an overjet greater than 2.5 mm ($n = 49$), a narrow mandible ($n = 69$), a high and narrow hard palate ($n = 56$), an elongated uvula and redundant soft tissue and/or enlarged tonsils in the oropharynx ($n = 47$), and temporomandibular joint laxity with or without a click ($n = 33$).

Investigation of the type of trauma suffered by the sleep-disordered breathing patients revealed that all 16 whiplash patients had sleep-disordered breathing. The identification of "abnormal breathing during sleep" based on the Sleep Disorders Questionnaire and clinical examination indicated sleep-disordered breathing.

Patients with sleep-disordered breathing had a mean ESS score of 12.8 ± 2.1 (range, 8 to 15) and a mean MSLT of 7.9 ± 2.2 minutes (range, 5.2 to 11 minutes). Nine patients with obstructive sleep apnea (OSA) had an MSLT between 10 and 11 minutes. The same nine patients all had OSA at polygraphic recording, and their mean AHI was 18 ± 7 events per hour of sleep. The mean Hamilton Depression Scale score for sleep-disordered breathing patients was 15 ± 4 (range, 10 to 20).

Evaluation of behavioral status before head trauma. For many of our patients, the association between the head-neck trauma and sleepiness was not strongly challenged, but this was not the case for patients diagnosed with sleep-disordered breathing. Although sleep-disordered breathing could be a consequence of head-neck trauma, in some patients it could have been a preexisting condition—one that may have even contributed to the accident. For this reason, we had to evaluate systematically (as much as possible) the health status of the patient before the accident. The following information was obtained:

- Sleep/wake history from bed partners
- Reports from coworkers on sleep/wake behavior before the accident
- Evaluation from employers, including absenteeism and accidents
- Financial information demonstrating the level of gainful employment for 2 years before the accident
- Health records from health agencies (for specific professions) and from private physicians, with patient authorization
- Driving records (with the patient's or lawyer's authorization)

Fifty-five of the 59 sleep-disordered breathing patients had a regular bed partner before and at the time of the head trauma. All had regular employment before the accident. Several patients had jobs that required regular health checkups and performance evaluations.

We were unable to demonstrate the presence of overt symptoms of sleep-disordered breathing in 52 of 59 patients. Six patients were identified as snorers before the accident (one patient unknown). In none of the patients were we able to demonstrate overt and regular sleepiness. All patients were fully employed and had no history of driving or professional accidents before the head trauma. There was no information indicating difficulty on the job, including personnel records at health agencies.

Evaluation of depression and sleepiness. The Hamilton Depression Scale score was taken as an index of depression. Daytime sleepiness was assessed by MSLT. The patient population studied consisted of 28 patients with an MSLT of 5 minutes, 58 patients with an MSLT of > 5

minutes and ≤ 10 minutes, and 15 patients with an MSLT of > 10 minutes. The corresponding mean Hamilton Depression Scale scores were, respectively, 19 ± 6 , 17 ± 7 , and 12 ± 6 . Analysis of variance did not indicate a significant difference between the two "sleepy" groups, but there was a difference in depression scores ($p < 0.05$) between the group with the worst sleepiness (MSLT of 5 minutes) and the group with an MSLT of > 10 minutes.

Presence of sleep-onset REM periods (SOREMP) at MSLT. SOREMPs at MSLT are an abnormal result that has been associated with narcolepsy. Five patients had two or more SOREMPs during the five MSLT naps.

Two of the 51 patients with MSLT scores of 5 minutes had two SOREMPs during the MSLT; four of these 51 patients had one SOREMP. Three of the 101 patients with MSLT scores between 5 and 10 minutes had SOREMPs; four of the 101 patients had one SOREMP. No SOREMP was noted in patients with MSLT scores of 10 minutes.

None of the patients with SOREMPs reported cataplexy. The two patients with an MSLT of 5 minutes and two SOREMPs were in the group with the shortest total sleep time (360 minutes) and complained of pain. One of the patients with an MSLT score between 5 and 10 minutes and two SOREMPs was an overweight patient with obstructive sleep apnea. Of the five patients with two SOREMPs, four had DR15 DQ6 (DR2 DQ1) HLA typing, and two of the four were positive for the haplotypes.⁹

Patients with an MSLT of 12 minutes. Based on MSLT results, eight patients were not "objectively sleepy."¹⁰ When first seen at the clinic, however, a mean of 13 ± 3 months after the head trauma, they were unable to perform activities of daily living due to sleepiness and were indifferent to daily requirements and to the task at hand. These patients could perform simple tasks if challenged, but if unchallenged their activity would cease and they would resume behavior that was indicative of sleepiness or sleep. These patients were otherwise well oriented and had appropriate recent memory. They were not confused, and had no speech defects. Although none of these patients had head fracture, brain hemorrhage, or immediate neurosurgical intervention, all had been in a coma (no information on stage) for a mean duration of 27 ± 10 hours. No focal lesions were noted on clinical evaluation, EEG, or MRI, and no seizures were noted during the follow-up period. Their mean ESS score was 19 ± 3 , and their mean Hamilton Depression Scale score was 14 ± 4 . Five of these nine patients had been given amphetamines that were reported to be helpful.

Polygraphic recordings of patients with an MSLT of 12 minutes. Patients free of medication for a minimum of 3 weeks were monitored for 48 hours. During the first 24 hours, patients had 510 minutes of nocturnal monitoring. After breakfast, five nap MSLTs were scheduled. At the end of the first 24 hours, patients were left to choose their own bedtime, wake-up time, and nap times for the following 24 hours.

On day 1, the mean total nocturnal sleep time was 471 ± 25 minutes, and the mean percentages of total sleep time were the following: stage 1 non-REM (NREM) sleep, 21%; stage 2 NREM sleep, 49%; stages 3 to 4 NREM sleep, 12.2%; and REM sleep, 17.5%. The mean MSLT scores for the five naps ranged from 13 ± 3 minutes to 17 ± 2.1 minutes.

On day 2, the total bedtime was 975 ± 48 minutes. The mean total nocturnal sleep time was 459 ± 38 minutes, with presence of all sleep stages and states. Patients spent an additional 139 ± 21 minutes of inactive time during the day, sitting on chairs or couches with their eyes closed. During these inactive periods, the EEG in the central leads (C_3/A_2 , C_4/A_1) was very monomorphic, with continuous recording of EEG patterns in the 7 to 9-Hz range (i.e., the EEG patterns indicated drowsiness without the development of the classic sleep stages). Mean total daytime sleep time (stages 1 and 2) was 31 ± 14 minutes.

Factors related to subjective or MSLT-defined sleepiness. Several statistical models were investigated. The clinical variables selected for multiple logistic regression (stepwise) were age, coma of 24 hours, head fracture (there was collinearity associated with neurosurgical intervention), neurologic sequelae, Hamilton Depression Scale score of 18, sleep-disordered breathing, whiplash, nocturnal pain, nocturnal headache, antiepileptic medication intake, BMI, and total time in bed per 24-hour period. These clinical variables were considered for several dependent variables indicative of daytime sleepiness.

When the dependent variable was an ESS score of 16, the following variables were found to be significant: coma of 24 hours ($p < 0.001$), head fracture (collinearity with neurosurgical intervention and neurologic sequelae, $p < 0.01$), total time in bed > 8.5 hours ($p < 0.05$), and age of 30 years ($p < 0.05$). The R^2 of the model was 0.43.

When the dependent variable was an MSLT score of 5 minutes, the significant variables were coma of 24 hours, head fracture, and total time in bed. Age was not significant in this model. The R^2 of the model was 0.40.

An MSLT of 5 minutes and an ESS score of 16 were not independent variables.

To understand daytime sleepiness better, we also considered a nocturnal sleep time of 360 minutes as a dependent variable. The following variables were significant: report of daily pain ($p < 0.01$), neurologic sequelae (this was collinear with head fracture, which was not in the model; $p < 0.05$), and a Hamilton Depression Scale score of 18 ($p < 0.05$). The R^2 of the model was 0.20.

When sleep-disordered breathing was the dependent variable, the following 12 clinical variables were tested: age, coma of 24 hours, whiplash, head fracture, Hamilton Depression Scale score, nocturnal headache, nocturnal pain, loud regular snoring, antiepileptic drug intake, BMI, MSLT score, and total time in bed per 24-hour period. Only two variables were significant: whiplash and an MSLT of 8 minutes. Interestingly, BMI was not a significant variable. The R^2 of the model was 0.21.

Discussion. Our population is very heterogeneous, which reflects the variety of problems associated with post-traumatic hypersomnia and the need for the wide array of tests involved in our evaluation. Our experience, however, has led us to add the Maintenance of Wakefulness Test (MWT) to our evaluation because it provides additional information on patients' alertness.¹¹ We have also added the Beck Depression Inventory because our findings suggested that patients with severe daytime impairment are susceptible to reactive depression. We also did not find the ESS to be very useful; we have recently

switched to the Karolinska Sleepiness Scale.¹² Finally, we regret not keeping more DNA samples from our patients. An increasing number of genes in the control of sleep/wakefulness and chronobiology is being identified, and future investigation may reveal whether certain individuals are more susceptible to developing daytime sleepiness after head trauma.

How do we deal with the medicolegal dilemma? First we assess patients for objective evidence of "hypersomnia" using the MSLT. There are controversies on the usage of the MSLT as the most objective test for evaluation of daytime somnolence. Some prefer to evaluate the frequency and duration of sleep episode either in a control environment, or in the midst of daily activity during continuous 24-hour polygraphic recordings. This type of testing was also used when we faced a very large discrepancy between MSLT results and subjective sleepiness scales (as seen with our patients with compulsive presleep behavior).

There are also controversies regarding the definition of the limit of normalcy with the MSLT. Although an MSLT between 10 and 12 minutes is not considered to be pathologic according to the guidelines, an MSLT of 11 minutes is not uncommon in association with a report of "daytime tiredness" in the presence of an abnormal respiratory disturbance index.

The MSLT was developed at Stanford for the recognition of narcolepsy, but its use has been extended to the recognition of excessive daytime sleepiness. Until the MSLT and MWT are studied on a very large group of "normal" patients in different age groups, we will not know the specificity and sensitivity of these tests. We feel that there are probably overlaps between normal and pathologic scores, depending on the study population and age group.

In our population, based on one MSLT, 24 patients scored between 10 and 12 minutes. These patients probably had daytime sleepiness, but this question was not clearly resolved by our study.

Second, we looked for an abnormal distribution of REM sleep during the daytime. This is an objective demonstration of sleep-related pathology. The presence of two or more SOREMPs during the MSLT suggest possible post-traumatic narcolepsy (based on the definition of narcolepsy in the International Classification of Sleep Disorders¹³). In our population, two or more SOREMPs were uncommon. New data suggest that SOREMPs may be related to significant nocturnal sleep reduction and fragmentation.^{14,15} This seems to be the case in at least three of our five patients with two or more SOREMPs. However, SOREMPs may be related to a genetic predisposition (with involvement of DQ beta 0702), without the presence of the classic narcoleptic tetrad.¹⁶ Additional DNA studies on patients with head trauma and hypersomnolence are needed.

When the presence of "hypersomnia" based on the most commonly used MSLT was established, we then looked at whether the head trauma was responsible

for the reported daytime sleepiness, and whether there were preexisting features that might have predisposed or amplified the current complaint. This was difficult because we had polygraphic documentation of sleep-related breathing disorders in 59 subjects. We then examined previous documentation of the patient's alertness, daytime performance, and employment record before the accident. We concluded that there was no indication of behavioral consequences of sleep-disordered breathing before the head-neck trauma in 52 of the 59 patients. Six of the seven remaining subjects had a history of nocturnal snoring, but did not demonstrate any breathing disorder.

Our search of a preexisting risk factor was based on clinical evaluation, using published information on the relationship between maxillomandibular development, orofacial dysmorphism, and sleep-disordered breathing. We believe that in the presence of sleep-disordered breathing after head trauma, one should look for the aforementioned anatomic risk factors.

Interestingly, all patients with a history of whiplash had sleep-disordered breathing. This is important because one may not associate neck trauma with the development of sleep-disordered breathing. This relationship between whiplash and sleep-disordered breathing was noted 15 years ago.² One patient was followed for several years after his head trauma, and he experienced a progressive regression of his sleep-disordered breathing over a 2-year period. We have not observed this in our current study. In view that with the presence of anatomic risk factors,^{17,18} cervical fusion and surgical intervention on cervical discopathies leads commonly to sleep-disordered breathing; the development of sleep-disordered breathing secondary to whiplash may not be surprising.

Our approach has been conservative when dealing with medicolegal patients. When patients with anatomic risk factors were found to have sleep-disordered breathing and clinical symptoms that date from the time of the trauma, we concluded that a "health problem occurred on a preexisting condition."

We considered also the relationship between the severity of the subjective sleepiness, the scores of the sleepiness scale, and the score of the MSLT. Patients who had head fracture and coma ≥ 24 hours had severe daytime sleepiness. These patients had other neurologic sequelae, including seizure disorder and antiepileptic medication intake. Although these medications might have worsened their daytime sleepiness, it is less likely because other patients on similar antiepileptic drugs had normal results on the MSLT.

This finding raised an obvious question. Why did we not see a greater number of patients with severe head trauma and daytime sleepiness? One possibility is that daytime sleepiness is an uncommon consequence of head trauma. Another possible explanation is that there are so many neurologic sequelae to head

trauma that sleepiness may not be judged to be a significant problem. It might have been treated symptomatically by the neurologist, or questions on daytime sleepiness, tiredness, or fatigue might not have been asked. Our investigations suggest the need to question systematically this impairment, which needs treatment and compensation if legally justified.

The logistic regression analysis further identified factors that seem obvious but are not necessarily taken into consideration when daytime sleepiness is involved. Pain and neurologic sequelae disrupted sleep and led to a decrease in total nocturnal sleep time for many of our patients. The treating neurologist must always consider the impact of a patient's handicap on sleep. The patients with other neurologic sequelae (mono-, bi-, tetraparesis) and history of lasting coma with or without head fracture were in the group of patients with an MSLT of ≤ 7 minutes and most commonly ≤ 5 minutes. The sleepiness of these patients may be related largely to their report of nocturnal distress and disrupted sleep. Any nocturnal disrupted sleep has an impact on the following daytime's wakefulness, and impairment of both will have an impact on the patient's rehabilitation. A full sleep/wake history must be obtained on the patient with head trauma and neurologic sequelae. Daytime sleepiness must be explored and not assumed to be related to palsy.

An important element of our investigation was also to note a disabling problem of "pseudohypersomnia," which we believe is related to the head trauma. We had eight patients who were apathetic and subjectively sleepy but had normal MSLT scores. Apathy and daytime sleepiness have been associated with bilateral paramedian thalamic infarcts. Most of these lesions also involved the adjacent hypothalamus. Ischemic infarcts in the paramedian region of the thalamus have been shown to be associated with subthalamic and mesencephalic lesions¹⁹ and normal wakefulness, but otherwise apathetic and unmotivated behavior after the first month. The authors stressed that the problem was more of an "attention deficit" than a lack of alertness.

Another study²⁰ dissociated "sleepiness" from "compulsive presleep behavior" when monitoring a 32-year-old woman with bilateral obstruction of the posterior cerebral arteries at their posterior origin, leading to hypodense lesions in the paramedian parts of the thalamus. After an initial coma, the patient was severely apathetic. A 24-hour ambulatory EEG showed "a normal 8-hour sleeping pattern" with normal waking EEG, including when she demonstrated "presleep behavior." We have also had three patients with bilateral paramedian thalamic lesions²¹ who developed a similar pattern of apathetic behavior to the one reported here.

Significant deviations from the normal behavior associated with sleep were described in the athalamic cat²²—whether it be presomnic, somnic, or postsomnic behavior—with the persistence of normal

sleep. It seems that lesions that involve the paramedian nuclei of the thalamus are associated with a strong apathetic syndrome. In our eight post-traumatic patients, no lesion was seen on MRI, but this does not eliminate the possibility of neurophysiological impairment secondary to the significant head trauma. However we have no clear substrate to explain the syndrome in these patients. Nonetheless, these patients had a significant neurologic handicap despite the absence of polygraphic findings supporting the diagnosis of hypersomnia. It is important to recognize this "pseudohypersomnia" and not to label patients as "malingerers."

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